

NATIONAL TEACHERS COLLEGE
School of Arts, Sciences, and Technology — BSIT Program

NTC_CC105 – DATA STRUCTURE W/LAB

WEEK 5 LABORATORY INSTRUCTION

Laboratory Activity 5

Instructor	Prof. Darwin C. Vargas, MIT
Program	BSIT
Academic Year	2026–2027
Programming Language	C++

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5. Laboratory Activity No.	Laboratory Activity 5
6. Laboratory Title	Linear Architecture Synthesis & Peer Code Audit

PART B: Baseline Unit Testing Suite Execution

Test Case ID	Target Function / Operation	Test Input Vector	Expected Output	Actual Output (Legacy Code)	Pass / Fail	Technical Remarks
TC- 01	add_item()	Add 10, 20, 30 (Normal)	Items added; Count = 3; Capacity = 5.	Item added successfully	Pass	Normal addition within initial capacity constraints works.
TC- 02	add_item()	Add 6 items (Triggers Resize)	Capacity doubles to 10; all items retained.	Items added, but old array memory leaked in the heap.	Fail	Dynamic allocation directly leaks the previous heap memory buffer.
TC- 03	printAll()	Display indices 0 to 4 clearly	Displays indices 0 to 4 cleanly.	Displays 5 valid items plus 1 garbage memory value.	Fail	Off-by-one loop condition ($i \leq \text{currentCount}$) reads past array boundary
TC- 04	findItem()	Return index 2	Returns index 2.	Returns index 2.	Pass	Target found correctly despite redundant inner loop execution overhead.
TC- 05	findItem()	Return -1	Returns -1.	Returns -1.	Pass	Non-existent element lookup handles the fallback return safely.

TC- 06	remove_item_at()	Item at index 2 removed; array shifts left; count = 4.	Item removed; array shifts left; count = 4.	Item removed; array shifts left correctly.	Pass	Operationally functional, but execution logic contains redundant shifting loops.
TC- 07	remove_item_at()	Rejects input with error message; no state change.	Rejects input with error; no state change.	Prints "Invalid index!", state unchanged.	Pass	Lower boundary checks catch negative index inputs properly.
TC- 08	remove_item_at()	Rejects input with error message; no state change.	Rejects input with error; no state change	Crash / Corrupts array state (Shifts invalid memory).	Fail	Lacks upper boundary validation checking (idx >= currentCount).
TC- 09	processMatrix()	Transposed grid output correctly formatted	Transposed grid output correctly formatted.	Displays transposed matrix.	Pass	Functions correctly but relies on hardcoded dimensions.
TC- 10	Memory Clean	Program Exit	Zero leaks reported by memory diagnostics.	Memory leak reported: lost heap allocation bytes.	Fail	Subsystem lacks a proper destructor cleanup delete[] call.